

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Logic Reasoning & Mapping Exercise

COURSE NAME: ARTIFICIAL INTELLIGENCE
AND EXPERT SYSTEMS

COURSE CODE: MLA01

COURSE OUTCOME COVERED

CO3: *Implement propositional and first-order logic using resolution, unification, and inference techniques for knowledge representation and reasoning.*(BTL 3)

Assessment Tool Weightage: 15%

Total Marks: 100

Time: 60 Mins

No. of Questions: 5

Annexure A

Logic Reasoning & Mapping Exercise

Code of Conduct:

I **RASHMITHA SENTHIL KUMAR (192525034)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criterion	Weightage	Marks Obtained
1. Problem Analysis and Knowledge Identification	15%	
2. Logical Representation and Mapping	20%	
3. Predicates, Variables, Constants and Quantifiers	15%	
4. Logical Connectives and Formula Construction	10%	
5. Inference and Logical Reasoning	15%	
6. Unification and Substitution	10%	
7. Resolution and Proof Construction	10%	
8. Mapping Diagram / Clarity and Presentation	5%	
Total	100%	

ANSWER ANY ONE OF THE FOLLOWING

Q.No	Use Case Scenario	
1	Smart Healthcare Diagnosis	A hospital uses an AI system to identify possible diseases from patient symptoms. Ravi has fever, cough, and body pain. Patients with fever and cough may have flu. Patients with flu and body pain may have viral infection. Patients with viral infection require medical attention. Tasks: (i) Identify facts, entities, predicates, variables, and constants. (ii) Represent the knowledge using Propositional Logic. (iii) Represent the knowledge using FOL with suitable quantifiers. (iv) Perform unification for two relevant predicates. (v) Apply resolution to prove whether Ravi has a viral infection. (vi) Show the complete inference sequence and final conclusion.
2	Student Academic Advising	A university AI system recommends academic support based on attendance and examination performance. Priya has attendance below 75% and has failed two courses. Students with low attendance require attendance counseling. Students who fail two or more courses require academic counseling. Students requiring both types of counseling need academic support. Tasks: (i) Identify the knowledge components. (ii) Convert the statements into Propositional Logic. (iii) Develop FOL representations using predicates and quantifiers. (iv) Identify suitable substitutions and perform unification. (v) Use resolution to prove whether Priya requires academic support. (vi) Explain the reasoning steps leading to the conclusion.
3	Intelligent Loan Approval	A bank wants to use an AI knowledge-based system for preliminary loan approval. Arun has stable employment, sufficient income, a high credit score, and good repayment history. Customers with stable employment and sufficient income qualify for preliminary approval. Customers with a high credit score and preliminary approval are classified as low-risk. Low-risk customers with good repayment history can be recommended for loan approval. Tasks: (i) Identify entities, facts, predicates, and rules. (ii) Map the scenario into Propositional Logic. (iii) Represent the rules using FOL.

		(iv) Demonstrate unification and substitution. (v) Use resolution to prove or disprove LoanApproved(Arun). (vi) Provide the final inference with justification.
4	Vehicle Fault Diagnosis	An automobile service center uses an expert system to diagnose vehicle faults. Car101 does not start and its headlights are dim. Vehicles that do not start and have dim headlights may have a battery problem. Vehicles with a battery problem require battery inspection. Vehicles requiring battery inspection are marked for service. Tasks: (i) Extract the facts and rules from the scenario. (ii) Represent them using Propositional Logic. (iii) Develop suitable FOL predicates and rules. (iv) Perform unification for the vehicle-related predicates. (v) Apply resolution to determine whether Car101 has a battery problem. (vi) Derive whether Car101 should be marked for service and explain the inference.
5	Cybersecurity Threat Detection	An organization uses an AI system to identify cyber threats. User101 has five failed login attempts and logged in from an unusual location. Multiple failed login attempts indicate a possible brute-force attack. A brute-force attack combined with an unusual login location indicates suspicious activity. Suspicious activity combined with unauthorized access to confidential files indicates a possible security breach. Tasks: (i) Identify the facts, entities, predicates, and rules. (ii) Map the knowledge into Propositional Logic. (iii) Represent the rules using FOL. (iv) Demonstrate unification and substitution. (v) Apply resolution to determine whether User101 is involved in a possible security breach. (vi) Present the complete reasoning chain and final inference.

QUESTION 4 - VEHICLE FAULT DIAGNOSIS

Aim:

To design a logic-based expert system for vehicle fault diagnosis by representing the given knowledge using Propositional Logic and First-Order Logic (FOL), applying unification and substitution, constructing a resolution proof, and deriving whether Car101 should finally be marked for service.

1. Problem Understanding and Knowledge Identification:

An automobile service center uses an expert system to diagnose faults from observable vehicle symptoms. In the given scenario, Car101 does not start and its headlights are dim. The domain knowledge states that the simultaneous occurrence of these two symptoms indicates a battery problem. A detected battery problem requires battery inspection, and a vehicle requiring battery inspection must be marked for service.

The reasoning objective is therefore twofold: first, to prove $\text{BatteryProblem}(\text{Car101})$; and second, to propagate that conclusion through the remaining rules until the final service decision $\text{MarkedForService}(\text{Car101})$ is obtained.

Table 1. Knowledge Components Identified from the Scenario

Component	Representation
Entity	Vehicle
Constant	Car101
Variable	x - represents any vehicle
Facts	$\text{DoesNotStart}(\text{Car101})$, $\text{DimHeadlights}(\text{Car101})$
Predicates	$\text{DoesNotStart}(x)$, $\text{DimHeadlights}(x)$, $\text{BatteryProblem}(x)$, $\text{BatteryInspection}(x)$, $\text{MarkedForService}(x)$
Rules	R1: symptoms imply battery problem; R2: battery problem implies inspection; R3: inspection implies service
Final goal	$\text{MarkedForService}(\text{Car101})$

2. Propositional Logic Representation:

For the specific vehicle Car101, the scenario can be mapped into propositions as follows:

P: Car101 does not start.

Q: Car101 has dim headlights.

R: Car101 has a battery problem.

S: Car101 requires battery inspection.

T: Car101 is marked for service.

The propositional knowledge base is:

Facts: P, Q

Rule 1: $(P \wedge Q) \rightarrow R$

Rule 2: $R \rightarrow S$

Rule 3: $S \rightarrow T$

Because P and Q are both true, Rule 1 yields R. Rule 2 then yields S, and Rule 3 finally yields T. Thus, the propositional mapping already establishes a complete path from the observed symptoms to the service decision.

3. First-Order Logic Representation:

To generalize the rules so that they can be used for any vehicle, let x denote an arbitrary vehicle and let Car101 denote the specific vehicle being diagnosed.

F1: DoesNotStart(Car101)

F2: DimHeadlights(Car101)

R1: $\forall x [(DoesNotStart(x) \wedge DimHeadlights(x)) \rightarrow BatteryProblem(x)]$

R2: $\forall x [BatteryProblem(x) \rightarrow BatteryInspection(x)]$

R3: $\forall x [BatteryInspection(x) \rightarrow MarkedForService(x)]$

The universal quantifier $\forall$ indicates that each diagnostic rule applies to every vehicle satisfying its antecedent. The conjunction $\wedge$ in R1 is essential because both symptoms must be present together. The implication $\rightarrow$ expresses the direction of the diagnostic reasoning.

4. Predicates, Variables, Constants, Quantifiers and Logical Connectives:

The predicate DoesNotStart(x) expresses a starting failure, while DimHeadlights(x) represents reduced headlight intensity. BatteryProblem(x), BatteryInspection(x), and MarkedForService(x) represent successive diagnostic and service states. The variable x permits general rules, whereas Car101 is a constant. The scenario uses conjunction ($\wedge$) and implication ($\rightarrow$). During resolution, implications are transformed into clauses containing disjunction ($\vee$) and negation ($\neg$). No equivalence or existential statement is asserted by the supplied scenario, so these are not introduced artificially.

5. Unification and Substitution:

Unification determines the substitution that makes a general predicate match the specific facts for Car101. The Most General Unifier (MGU) required here is:

$$\theta = \{x / \text{Car101}\}$$

Relevant unification steps are:

- DoesNotStart(x) and DoesNotStart(Car101) $\Rightarrow \{x / \text{Car101}\}$
- DimHeadlights(x) and DimHeadlights(Car101) $\Rightarrow \{x / \text{Car101}\}$
- BatteryProblem(x) and BatteryProblem(Car101) $\Rightarrow \{x / \text{Car101}\}$

After applying θ to R1, the instantiated rule becomes:

$$\text{DoesNotStart(Car101)} \wedge \text{DimHeadlights(Car101)} \rightarrow \text{BatteryProblem(Car101)}$$

Both antecedents are supplied facts, so the instantiated rule is satisfied and BatteryProblem(Car101) is derived.

6. Resolution Proof for BatteryProblem(Car101):

Resolution is used to prove the query BatteryProblem(Car101) by contradiction. The implication in R1 is first converted to clause form, and the query is negated.

C1: DoesNotStart(Car101)

C2: DimHeadlights(Car101)

C3: $\neg \text{DoesNotStart}(x) \vee \neg \text{DimHeadlights}(x) \vee \text{BatteryProblem}(x)$

C4: $\neg \text{BatteryProblem(Car101)}$ [Negated query]

Apply $\theta = \{x / \text{Car101}\}$ and resolve C3 with C4:

C5: $\neg \text{DoesNotStart}(\text{Car101}) \vee \neg \text{DimHeadlights}(\text{Car101})$

Resolve C5 with C1:

C6: $\neg \text{DimHeadlights}(\text{Car101})$

Resolve C6 with C2:

EMPTY CLAUSE

The empty clause indicates contradiction. Therefore, the negation of the query cannot hold, and $\text{BatteryProblem}(\text{Car101})$ is logically proved.

7. Complete Inference and Logical Reasoning:

After proving the battery problem, the remaining rules provide a direct inference sequence:

Step 1: $\text{DoesNotStart}(\text{Car101}) \wedge \text{DimHeadlights}(\text{Car101})$

Step 2: $R1 \Rightarrow \text{BatteryProblem}(\text{Car101})$

Step 3: $R2 \Rightarrow \text{BatteryInspection}(\text{Car101})$

Step 4: $R3 \Rightarrow \text{MarkedForService}(\text{Car101})$

Each conclusion becomes the premise for the next rule. There is no unsupported intermediate assumption: every step follows either from a supplied fact or from a supplied rule. Hence the reasoning chain is complete and traceable.

8. Mapping Diagram / Logic Flow:

The mapping diagram illustrates the complete reasoning process used by the Vehicle Fault Diagnosis expert system. The system begins with the observed facts that Car101 does not start and has dim headlights. These facts are matched with the logical rules using the substitution $\{x / \text{Car101}\}$. Rule R1 infers a battery problem, Rule R2 determines that battery inspection is required, and Rule R3 concludes that the vehicle must be marked for service. Thus, the diagram provides a clear visual mapping from the initial symptoms to the final service decision.

Inference Flow:

$$\begin{aligned} &\text{DoesNotStart}(\text{Car101}) + \text{DimHeadlights}(\text{Car101}) \rightarrow \text{BatteryProblem}(\text{Car101}) \rightarrow \\ &\quad \text{BatteryInspection}(\text{Car101}) \rightarrow \text{MarkedForService}(\text{Car101}) \end{aligned}$$

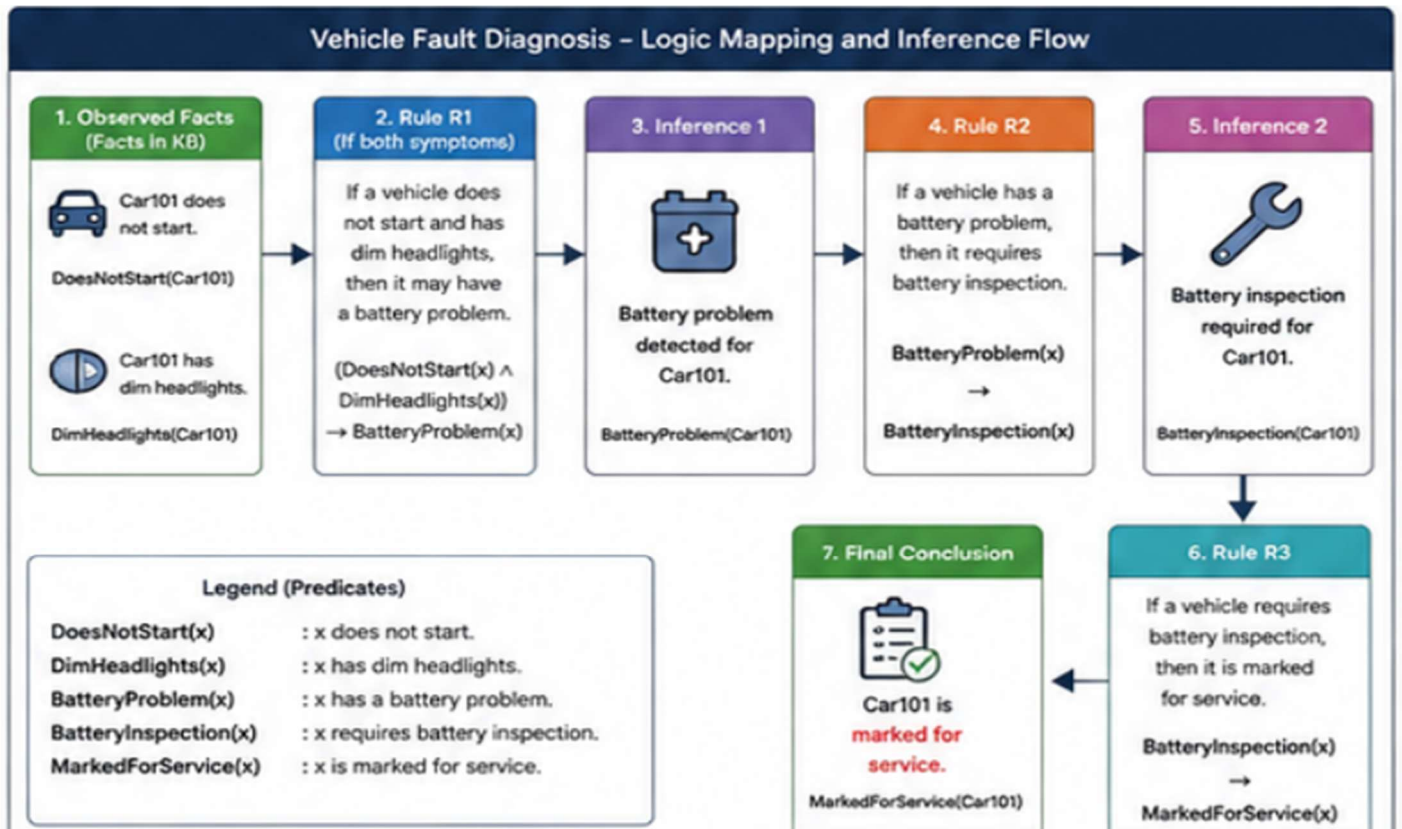


Fig. 1. Vehicle Fault Diagnosis - Logic Mapping and Inference Flow

9. Final Inference and Result

The resolution proof establishes $\text{BatteryProblem}(\text{Car101})$. Rule R2 then establishes $\text{BatteryInspection}(\text{Car101})$, and Rule R3 establishes $\text{MarkedForService}(\text{Car101})$. Therefore, Car101 should be marked for service because its observed symptoms satisfy the battery-problem rule and the resulting diagnosis activates the inspection and service rules.

10. Conclusion

The Vehicle Fault Diagnosis exercise demonstrates accurate mapping of a real-world service-center scenario into symbolic logic. Propositional Logic captures the specific Car101 case, while FOL generalizes the knowledge for any vehicle. The MGU $\{x/\text{Car101}\}$ correctly applies the rules to the observed vehicle, resolution formally proves the battery-problem query, and the inference chain leads to the final conclusion $\text{MarkedForService}(\text{Car101})$. The result is fully supported by the facts and rules provided in Question 4.

EVALUATION RUBRICS:

Criterion	Weightage (%)	Excellent	Good	Average	Poor
Problem Understanding & Knowledge Identification	15%	13–15% – Clearly identifies entities, facts, relationships, assumptions, and reasoning requirements from the given scenario.	10–12% – Identifies most relevant knowledge with minor omissions.	7–9% – Identifies basic knowledge but misses several important elements.	0–6% – Poor or incorrect identification of the given knowledge.
Logical Representation & Mapping	20%	17–20% – Accurately maps real-world statements into appropriate propositional and first-order logic representations.	13–16% – Provides mostly correct logical mappings with minor errors.	9–12% – Provides partial mappings with several logical errors.	0–8% – Logical mapping is largely incorrect or missing.
Predicates, Variables, Constants & Quantifiers	15%	13–15% – Correctly identifies and applies predicates, variables, constants, universal and existential quantifiers.	10–12% – Uses the concepts correctly with minor errors.	7–9% – Demonstrates partial understanding with some incorrect usage.	0–6% – Concepts are incorrectly applied or missing.
Logical Connectives & Formula Construction	10%	9–10% – Correctly applies AND, OR, NOT, implication, and equivalence to	7–8% – Mostly correct use of connectives with minor errors.	5–6% – Basic use of connectives but several errors occur.	0–4% – Incorrect or incomplete formula construction.

		construct valid logical formulas.			
Inference & Logical Reasoning	15%	13–15% – Performs accurate step-by-step reasoning and derives valid conclusions from the given knowledge.	10–12% – Reasoning is mostly correct with minor gaps.	7–9% – Provides partial reasoning with some incorrect conclusions.	0–6% – Reasoning is incorrect, unsupported, or missing.
Unification & Substitution	10%	9–10% – Correctly identifies substitutions and Most General Unifier (MGU) and applies them accurately.	7–8% – Correctly performs most unification steps with minor errors.	5–6% – Demonstrates basic understanding but has several errors.	0–4% – Unable to perform unification correctly.
Resolution / Proof Construction	10%	9–10% – Correctly applies resolution and presents a clear, logically valid proof of the query.	7–8% – Resolution is mostly correct with minor errors.	5–6% – Shows partial understanding but proof contains gaps.	0–4% – Resolution is incorrect or not demonstrated.
Clarity, Mapping Diagram & Presentation	5%	5% – Provides a clear, well-organized mapping/logic diagram with accurate notation and excellent presentation.	4% – Presentation is clear with minor formatting or diagram issues.	3% – Basic presentation with limited clarity.	0–2% – Poorly organized, unclear, or incomplete presentation.